

24-10-2024
22:00

ASSIGNMENT

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Operating Systems

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1)

Given:

- **Processes:** P1, P2, P3, P4, P5
- **Burst Time:** 2 ms, 1 ms, 8 ms, 4 ms, 5 ms
- **Priority:** 2, 1, 4, 2, 3
- **Arrival Time:** 0 ms (for all processes)

a. Gantt Charts for FCFS, SJF, Nonpreemptive Priority, and RR (Quantum = 2)

1. **FCFS (First-Come, First-Served)**

- Execution is based on the order of arrival.
- Order: P1 → P2 → P3 → P4 → P5

Process Start Time End Time

P1	0	2
P2	2	3
P3	3	11
P4	11	15
P5	15	20

2. **Gantt Chart:**

3. Copy code

4. | P1 | P2 | P3 | P4 | P5 |

5. 0 2 3 11 15 20

6. **SJF (Shortest Job First)**

- Shortest Burst Time is selected first.
- Order: P2 → P1 → P4 → P5 → P3

Process Start Time End Time

P2	0	1
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	Process	Start Time	End Time
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P1	1	3
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P4	3	7
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P5	7	12
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P3	12	20
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7. **Gantt Chart:**

8. Copy code

9. | P2 | P1 | P4 | P5 | P3 |

10. 0 1 3 7 12 20

11. **Nonpreemptive Priority (Lower number means higher priority)**

- Higher priority is chosen first.
- Order: P2 → P1 → P4 → P5 → P3

	Process	Start Time	End Time
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P2	0	1
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P1	1	3
----	---	---

P4	3	7
----	---	---

P5	7	12
----	---	----

P3	12	20
----	----	----

12. **Gantt Chart:**

13. Copy code

14. | P2 | P1 | P4 | P5 | P3 |

15. 0 1 3 7 12 20

16. **Round Robin (Quantum = 2)**

- Processes are executed for a maximum of 2 ms at a time in a round-robin fashion.

First cycle:

- P1 executes from 0 to 2.
- P2 executes from 2 to 3 (completes).
- P3 executes from 3 to 5.
- P4 executes from 5 to 7.

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- P5 executes from 7 to 9.

Second cycle:

- P3 executes from 9 to 11.
- P4 executes from 11 to 13.
- P5 executes from 13 to 15.

Third cycle:

- P3 executes from 15 to 17.
- P5 executes from 17 to 19.

Fourth cycle:

- P3 executes from 19 to 20 (completes).

Gantt Chart:

Copy code

```
| P1 | P2 | P3 | P4 | P5 | P3 | P4 | P5 | P3 | P5 | P3 |  
0   2   3   5   7   9   11  13  15  17  19  20
```

b. Turnaround Time (TAT) for Each Algorithm

Turnaround time = Completion time - Arrival time

Process FCFS SJF Priority RR

P1	2	3	3	2
P2	3	1	1	3
P3	20	20	20	20
P4	15	7	7	13
P5	20	12	12	19

c. Waiting Time (WT) for Each Algorithm

Waiting time = Turnaround time - Burst time

Process FCFS SJF Priority RR

P1	0	1	1	0
P2	2	0	0	2

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Process FCFS SJF Priority RR

P3	12	12	12	12
P4	11	3	3	9
P5	15	7	7	14

d. Minimum Average Waiting Time

1. FCFS:

- Average WT = $(0 + 2 + 12 + 11 + 15) / 5 = 40 / 5 = 8 \text{ ms}$

2. SJF:

- Average WT = $(1 + 0 + 12 + 3 + 7) / 5 = 23 / 5 = 4.6 \text{ ms}$

3. Nonpreemptive Priority:

- Average WT = $(1 + 0 + 12 + 3 + 7) / 5 = 23 / 5 = 4.6 \text{ ms}$

4. Round Robin:

- Average WT = $(0 + 2 + 12 + 9 + 14) / 5 = 37 / 5 = 7.4 \text{ ms}$

Conclusion:

- **SJF and Nonpreemptive Priority** result in the minimum average waiting time of **4.6 ms**.